

Parallax

A dispersion barbell: short rich single-name volatility, long cheap index convexity — fully autonomous, fully flat by the deadline.

Index vol is cheap. Single-name vol is rich.

- Late August 2026: VIX at 2026 lows, implied correlation in the bottom 1% since 2005.
- Single-stock volatility near the 98th percentile — index cheap, single-name rich.
- **dispersion_score** — vega-weighted single-name IV over index IV — computed every cycle and used as a hard 1.15 floor on Sleeve A sizing.
- Classic dispersion delta-hedges a full basket against the index; not viable on a 15-minute-delayed feed, so Parallax adapts the signal into outright defined-risk spreads.

Two sleeves, opposite signs on volatility

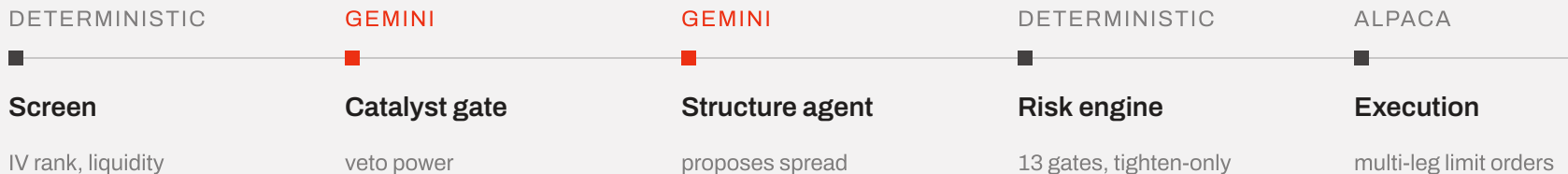
Sleeve A — Carry

Short idiosyncratic vol via \$5-wide put credit spreads / iron condors on liquid single names — IV rank > 50, IV30/HV20 > 1.15, 20–25 Δ short strike, 3–7 DTE. Capped at 0.8% NAV risk per name.

Sleeve B — Convexity

Long index vol via an SPY put debit spread (buy ~30 Δ / sell ~10 Δ), sized 1.4% NAV base (2.8% if NAV has grown), held through the Sep 4 NFP print and monetized into the open.

Screen → gate → propose → constrain → execute



Two Gemini calls (red), each with one narrow job, sit inside a fully deterministic screen and risk engine — schema-validated JSON in, never a silent pass-through.

AI proposes, code disposes

Catalyst gate

Vetoes rich-vol candidates with an unpriced binary catalyst pending. An error or malformed response is treated as `catalyst_risk=True` — a veto, never a silent approval.

Structure agent

Proposes strikes, expiry and limit price as schema-validated JSON. Sleeve B skips this gate by design — it exists to hold through a scheduled macro catalyst, not dodge one.

13 gates. Fixed order. Tighten-only.

Sizing & caps

Per-position and portfolio loss caps, concentration limits, and an -8% NAV drawdown kill switch that persists across restarts.

Structure & data

Defined-risk-only (no naked shorts), quote-staleness and liquidity floors, and the dispersion-score floor gating Sleeve A.

Timing & integrity

Earnings blackout, pre-NFP flatten, expiry-past-deadline, broker-reconciliation-is-truth, basket-capital-reservation.

- The engine can only RESIZE down or VETO — never enlarge a proposal — enforced by min() composition and a property test across 200+ randomized inputs.

Two gaps closed by design, not convention

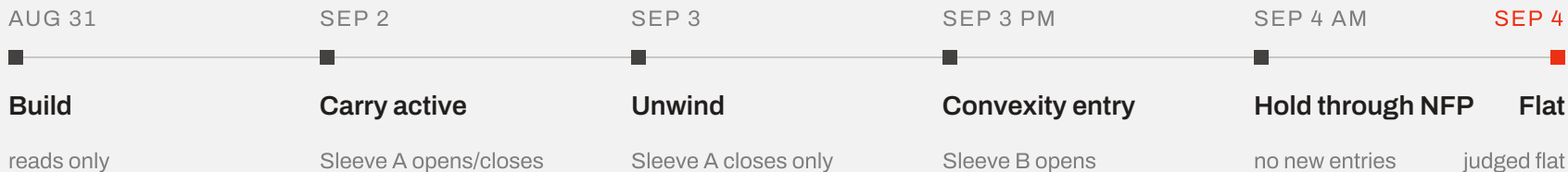
Broker-reconciliation-is-truth

Every cycle — and after every basket leg fill — broker state is diffed against the journal. A divergence halts new entries and logs CRITICAL, never trusting the agent's own internal record.

Sequential, capital-reserved baskets

Alpaca guarantees atomic fills only within one underlying, never across underlyings. A basket reserves its full max-loss up front, enters one name at a time, and re-runs the risk engine after each fill.

A dated state machine, not a model, decides



`gate_pre_nfp_flatten` and `gate_expiry_past_deadline` enforce these boundaries at the execution layer too — the account is guaranteed flat before judging regardless of model behavior.

The dispersion score board

- Every cycle's dispersion_score is now persisted to the journal — not just read live — so the thesis signal has a history, not a snapshot.
- The dashboard tracks it against the 1.15 floor alongside NAV, phase, and kill-switch state — one screen for the whole demo.
- Same signal, one source of truth: the gate that vetoes Sleeve A entries reads the row the board plots.

Verified against the real paper account

- One module imports alpaca-py — broker/alpaca_client.py — everything else calls through it, fully mockable.
- 180+ tests, ~95% coverage on risk/, zero live calls in CI.
- verify_day1.py caught two real bugs before they could fail live: mleg orders reject the symbol field, and Alpaca's debit/credit limit-price sign is inverted from this codebase's convention.
- Options orders are limit-only, enforced in code — not just config.

Thank you

Watch it trade: barbell run-cycle, the live dashboard, and an append-only journal — every decision, pass or reject, logged with its reason.